

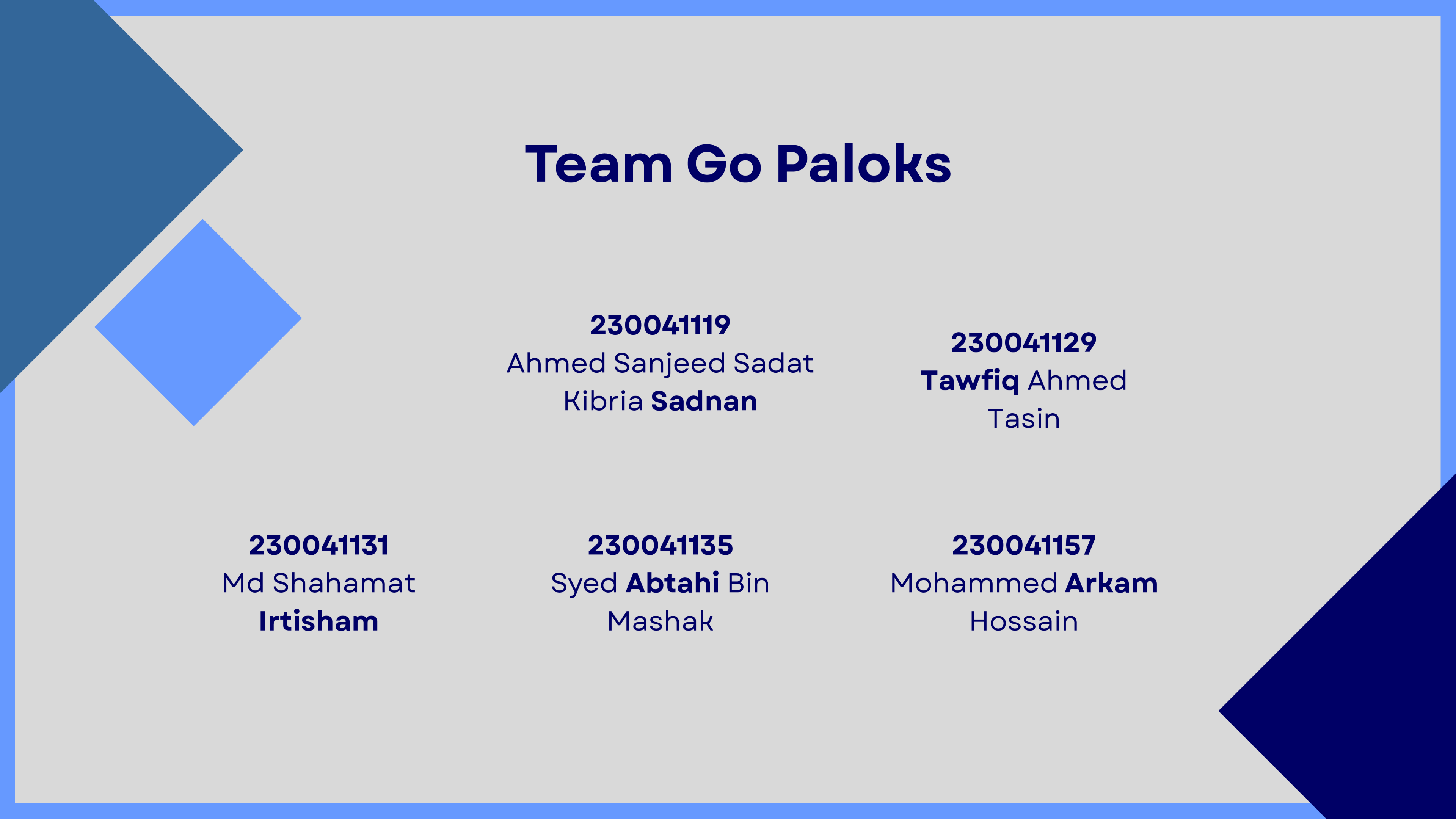
CSE 4408
SYSTEM ANALYSIS AND DESIGN LAB

Team: Go Paloks

Farm Management System

Presentation: 01

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Introduction

The System

We are planning to design an integrated system for a farm including cattle, chicken and fish farming with a sustainable approach.

Primary Goal

The goal is to build an data driven management system that is efficient and sustainable for the farm operations.



Organizational Context

Stakeholders

Farm Owner
Workers
Veterinarians
Suppliers
Customers

Environment

The environment of the system is semi digital and mostly manual. It is mainly resource constrained.

Existing Problems

**Manual Record
Keeping**

**No Centralized
System**

**Poor Tracking of
Feed Usage and
Production**

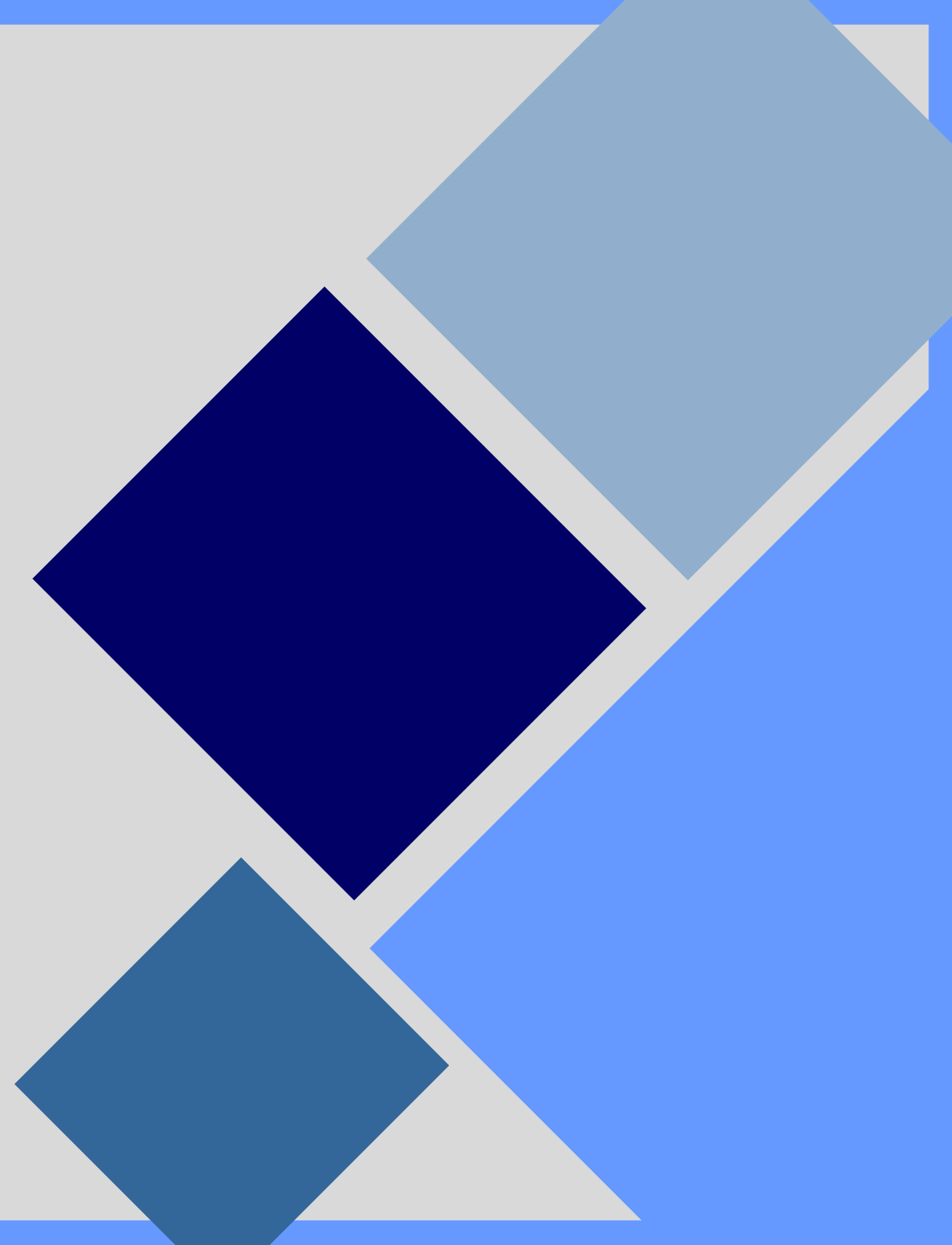
**Delayed
Decisions**

Need for SAD

System Analysis and Design (SAD) simplifies complex workflows to ensure the correct problems are solved from the start. By providing a clear blueprint, it increases efficiency and creates a structured system tailored to specific goals.

How a Systems Analyst can help

- Can make a bridge between the manual operated and IT system.
- Can analyze the needs of all the stakeholders as well as the system itself.
- Can design and develop the system according to the analyzed needs.



Opportunities

**Optimized
Resource Planning**

**Centralized
System**

**Real Time
Monitoring**

**Integrated Farm
Operations**

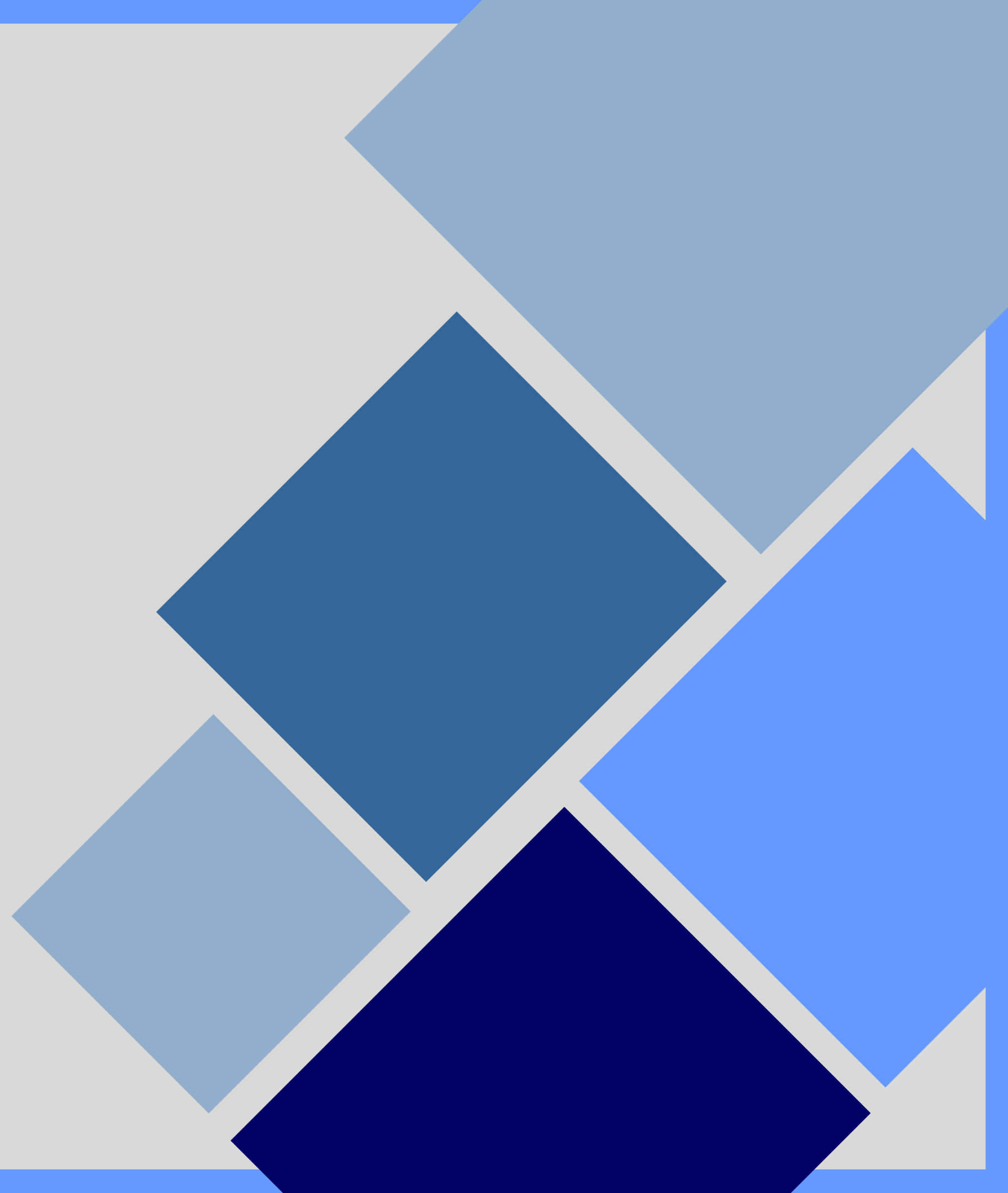
**Data Driven
Decisions**

Development Approach

We have considered the following approaches:

- SDLC
- Agile
- Object Oriented

And selected **Agile** as the primary methodology and **Object Oriented** as the design approach



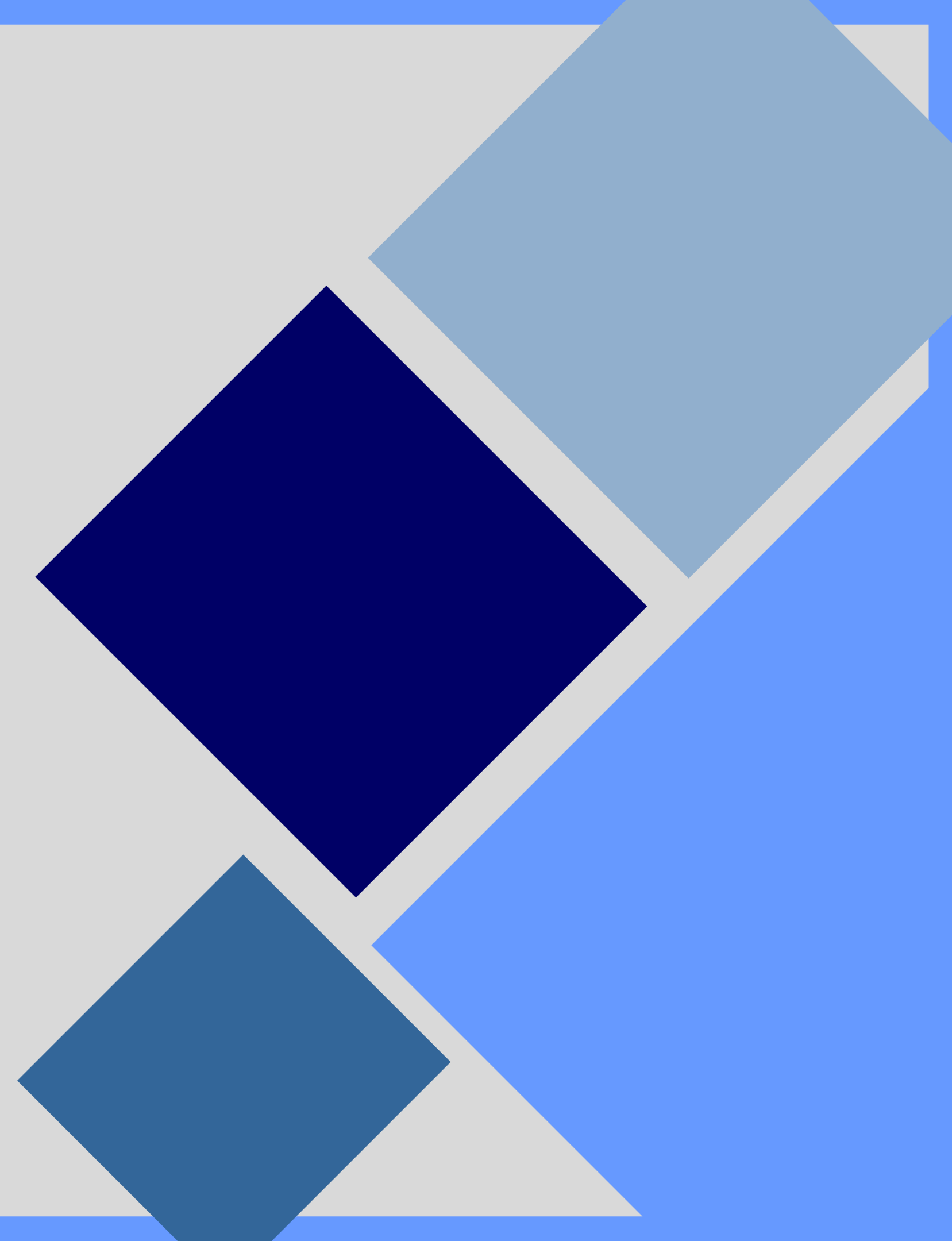
Why Agile?

The system will be built in phases. For instance:

- Phase 01: Cattle Farm
- Phase 02: Poultry Farm
- Phase 03: Fishery

In every phase the requirements will evolve. For this we need continuous development and feedback feature in our development methodology.

The approach is like - we will start small, learn, then expand.

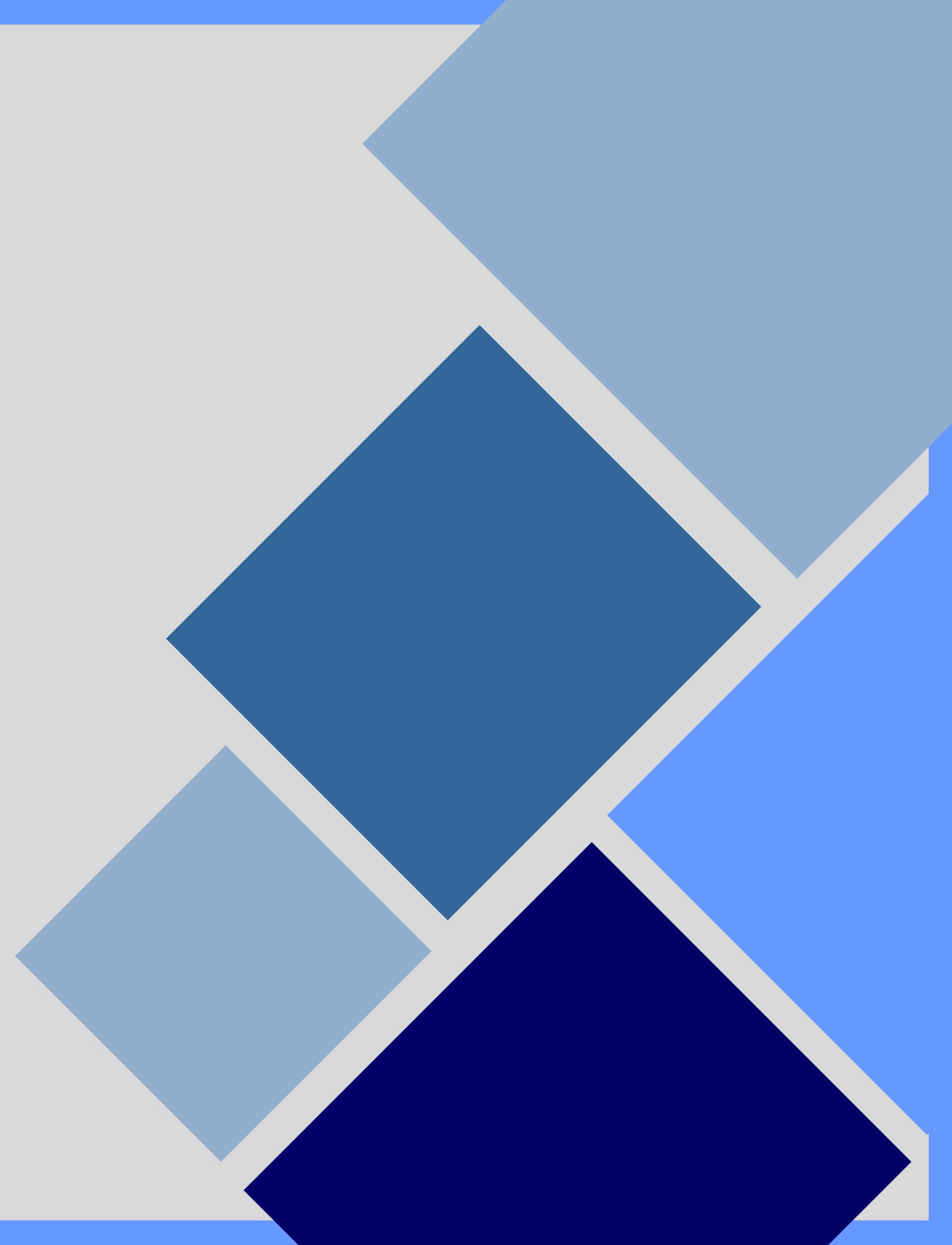


Why Object Oriented?

The farm entities are real world objects like: Cow, Chicken, Fish, Worker.

So, for the design approach we can use the object oriented approach. Thus we will get the benefit of modularity, reusability and easy expansion.

For example: Cow and Chicken both have feeding, tracking etc. So we can reuse the logic here.

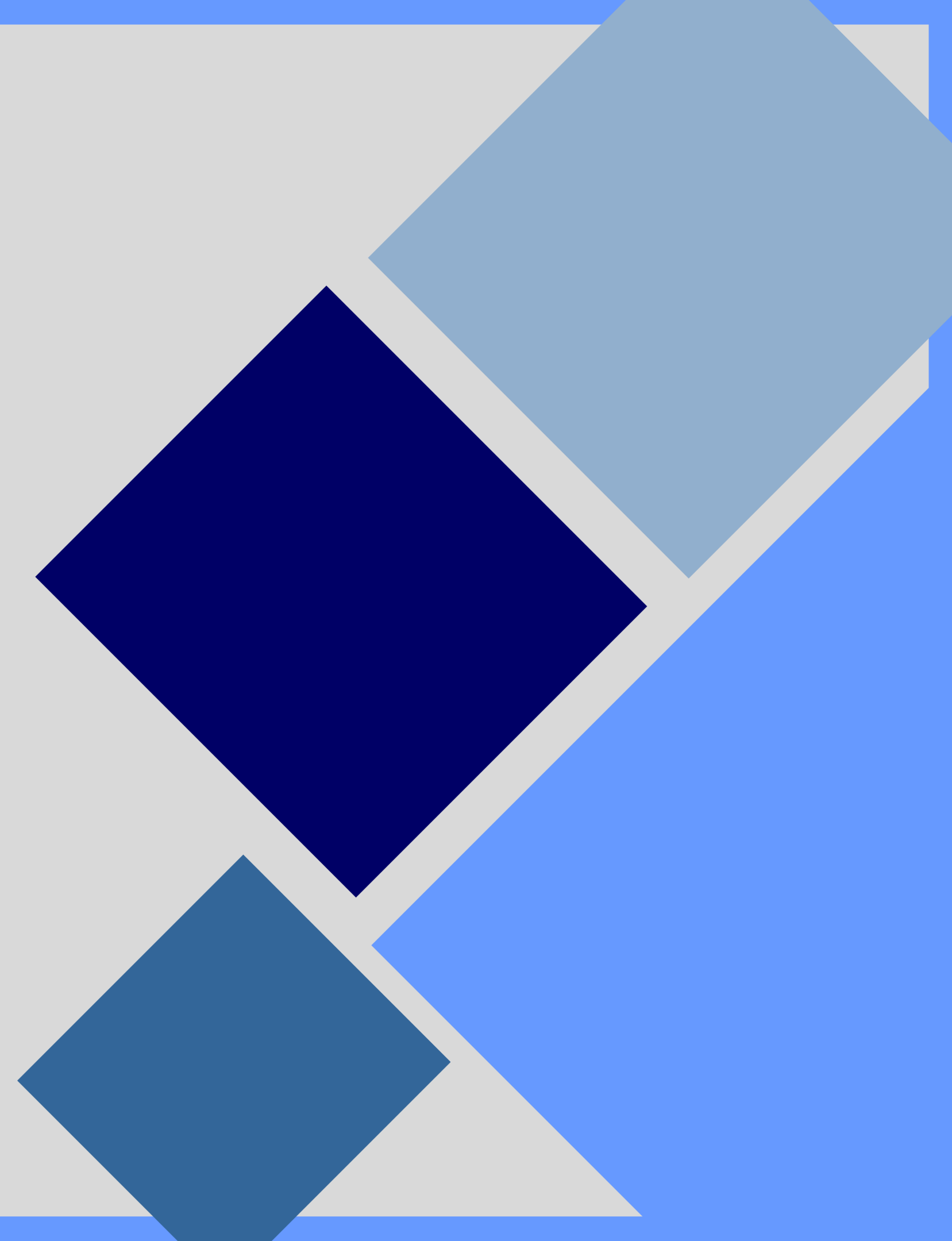


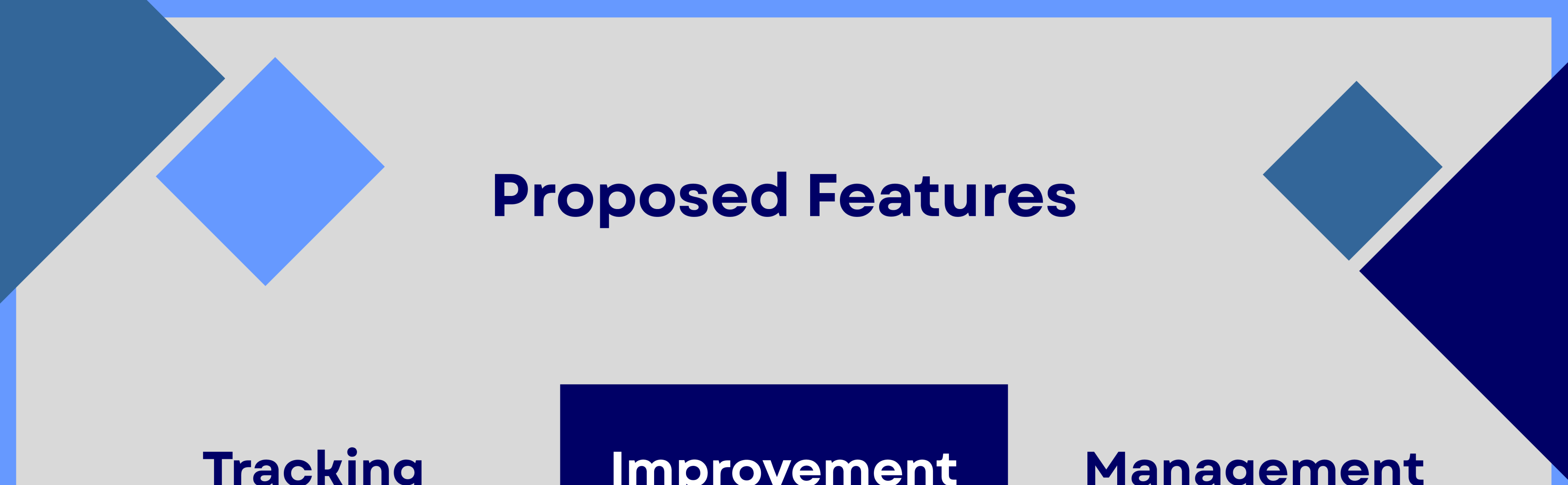
Why Not Waterfall?

We will not use Waterfall model because:

- The SDLC Waterfall model is too rigid and sequential.
- It will be hard to adapt to changing farm needs.
- Requires complete requirements upfront.

This approach is not suitable mainly because farm operations evolve continuously.





Proposed Features

Tracking

- Animal Health Tracking
- Production Tracing
- Overall System Tracking
- Income - Expense Tracking

Improvement

- Detailed Reports
- Visual Analytics
- Realtime Monitoring
- Data Driven Decisions

Management

- Inventory Management
- Worker Management
- Feed Management
- Waste Management

The background features a light grey central rectangle. Surrounding this rectangle are several geometric shapes in various shades of blue. In the top-left corner, there is a dark blue square and a medium blue triangle. In the top-right corner, there is a medium blue square and a light blue triangle. In the bottom-left corner, there is a light blue triangle. In the bottom-right corner, there is a dark blue triangle. The overall design is modern and minimalist.

QnA Session

Thank You !